



Figure 1: First screen of ElasticSearch output in the browser

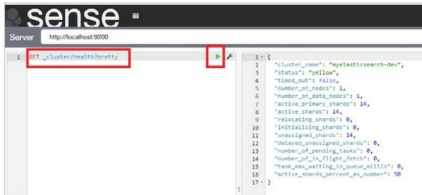


Figure 2: Using the Sense plugin for the ElasticSearch RESTful API

Installation and configuration

ElasticSearch needs a recent version of Oracle JVM, i.e., JDK 1.7U55, or higher. For more details, refer to https://www.elastic.co/support/matrix#show_jvm.

ElasticSearch can be downloaded from <https://www.elastic.co/downloads/elasticsearch>. It comes in a variety of formats—Zip, tar, DEB and RPM. DEB and RPM are packages for Linux based systems. We will go with the Zip format. Download the Zip file and extract it.

In the extracted folder, locate the config folder and open *elasticsearch.yml* in editing mode. In Linux based systems, simply right click and open the file using Gedit or a similar editor. On Windows, use Notepad or some other text editor.

In the *elasticsearch.yml* file, let us set some basic information:

- Find the line *cluster.name: my-application*. Remove the # sign so as to uncomment the line. Replace

my-application with some other more meaningful name.

- Find the line *node.name: node-1*, uncomment it by removing # at the start of the line and replace *node-1* with some meaningful name.
- Find the *network.host: 192.168.0.1*, uncomment it and instead of the IP address, specify 127.0.0.1.
- Save and close the file.

To run ElasticSearch, open the command prompt or terminal, and navigate to the *bin* directory. For Windows, it has batch files, and for Linux it has *sh* files.

- On a Windows system, execute *elasticsearch.bat* on the command prompt.
- On a Linux system, execute *.elasticsearch* in a terminal.
- Open the browser address 127.0.0.1:9200, as shown in Figure 1.

It shows that ElasticSearch is working. We can see the node name and the cluster name that we specified in *elasticsearch.yml* file.

This can also be tested using the *curl* command, which is available on Linux systems by default. For Windows, download *curl.exe* and add the same to the path. We don't require it for this article – it's been mentioned just for better understanding.

Open the terminal and run the following commands, one by one:

```
curl http://127.0.0.1:9200/
curl http://127.0.0.1:9200/?pretty
```

Using *curl* is not very comfortable, especially for beginners. We need a tool to visually interact with the ElasticSearch RESTful APIs. In the end, we would like to visualise the data also.

To do so in ElasticSearch, Kibana is used along with Sense, which is a Kibana plugin that allows a visual interaction with the RESTful API of ElasticSearch.

Download Kibana from <https://www.elastic.co/downloads/Kibana>. Download the appropriate versions of Windows or Linux. Extract the downloaded Zip file.

Open the command prompt or terminal and navigate to the Kibana folder. On a Linux machine, execute the following command:

```
./bin/Kibana --install elastic/sense
```

On a Windows machine, navigate to the *bin* folder inside the extracted Kibana folder and execute the following command:

```
kibana.bat --install elastic/sense
```

This will install the Sense plugin from the Web.

We need to start Kibana by running *kibana.bat* on Windows and *.kibana* on a Linux machine — both files can be found in the *bin* directory.

In a browser, open a new tab and in the address bar, type <http://127.0.0.1:5601> and press Enter. This will display the Kibana dashboard. We can configure it in a later section [i.e., visualising the data].